

Title : Differential Entropy

Category : Analysis

1- Objective :

This method measures the complexity and variability of EEG signals for each channel. It is used to show how random or irregular the brain activity is, and to compare EEG patterns between trials, labels, or conditions. Higher differential entropy means the signal is more variable and less predictable.

2- How it works :

The method works trial by trial. For each trial, it takes the EEG channels and computes the variance of each channel signal. Then it calculates differential entropy using:

$$H = 1/2 \log(2 \pi e \sigma^2)$$

The plot :

If there are at least two labels, the figure may also include:

- a topographic map for each label
- a difference map between labels (first two labels)
- a heatmap of channel vs. label
- a box plot showing trial-by-trial entropy distribution

If label information is not available, it shows:

- one overall topographic map
- a bar chart of the top channels by entropy

3- Example :

Suppose one trial has these channel values:

Fp1: variance = 4.0

Fp2: variance = 1.0

Cz: variance = 9.0

Then the entropy values will be:

$H_{Fp1} = 1/2 \log(2 \pi e * 4.0)$

$H_{Fp2} = 1/2 \log(2 \pi e * 1.0)$

$H_{Cz} = 1/2 \log(2 \pi e * 9.0)$

Since Cz has the highest variance, it will also have the highest entropy. On the topographic map, Cz will appear with a stronger color than the lower-variance channels.